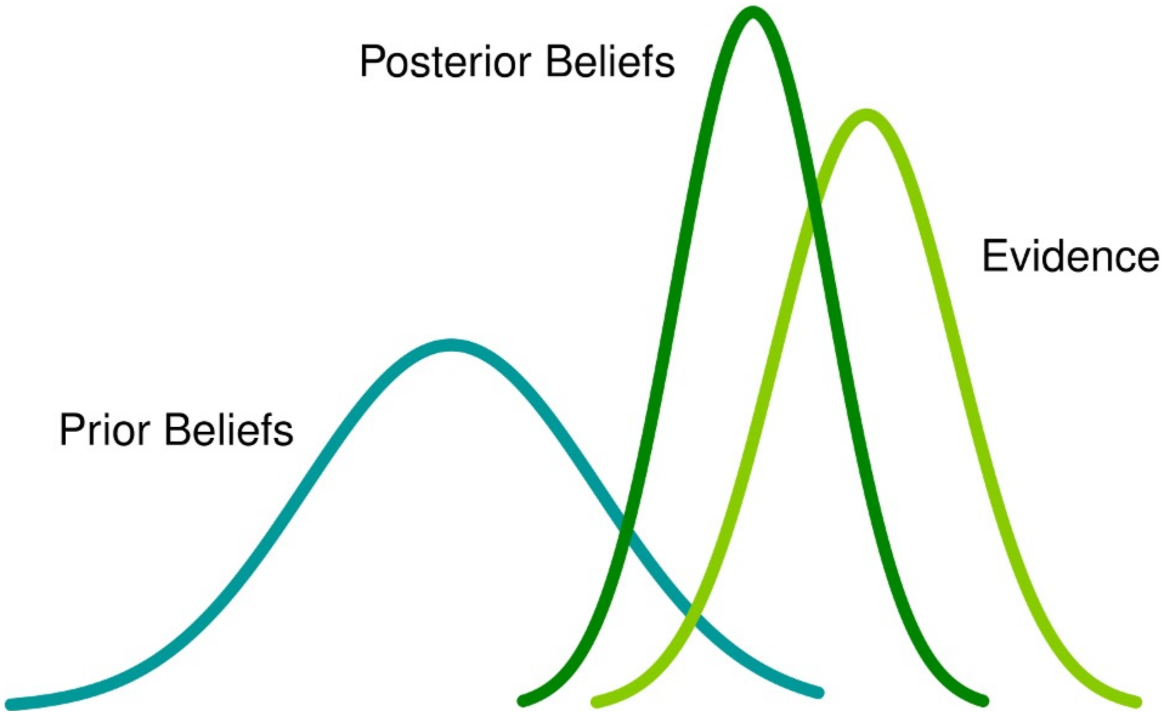
Lesson 7: Bayesian classifier



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贝叶斯决策论(Bayesian decision theory)

模型类型	核心思想
线性分类器	学习一个将样本分开的超平面 $f(x) = w^\top x + b$
SVM	最大化类别间的“间隔”，学习最具区分性的边界
决策树	通过特征划分不断细化空间，直到叶节点是“纯”的（属于某个类别）

虽然模型方法不同，但它们都要解决一个本质问题：

面对一个样本 **x**，我们该把它分到哪个类别，才能“最少犯错”？

- 所以无论是超平面、树、还是得分函数，
- 它们本质上都在尝试：
“最小化错误风险” 或 “最大化正确性”

贝叶斯决策论 (Bayesian decision theory)

概率框架下实施决策的基本理论

给定 N 个类别，令 λ_{ij} 代表将第 j 类样本误分类为第 i 类所产生的损失，则基于后验概率将样本 x 分到第 i 类的条件风险为：

$$R(c_i \mid \boldsymbol{x}) = \sum_{j=1}^{N} \lambda_{ij} P(c_j \mid \boldsymbol{x})$$

贝叶斯判定准则 **(Bayes decision rule)**：

$$h^*(\boldsymbol{x}) = \underset{c \in \mathcal{Y}}{\arg\min}\, R(c \mid \boldsymbol{x})$$

- h^* 称为 贝叶斯最优分类器**(Bayes optimal classifier)**，其总体风险称为 贝叶斯风险 **(Bayes risk)**
- 反映了 学习性能的理论上限

$h^*(x)$=arg max P(c|x)

判别式模型

贝叶斯决策论，不是一种模型，而是一种原则：
只要你能得到后验概率 **P(y|x)**，就能做最优分类。

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这些方法都是直接建模后验概率 **P(y|x)** 或者学习一个决策边界，从数据中“判别”出类别 —— 我们称这类方法为判别式模型 **(discriminative models)**

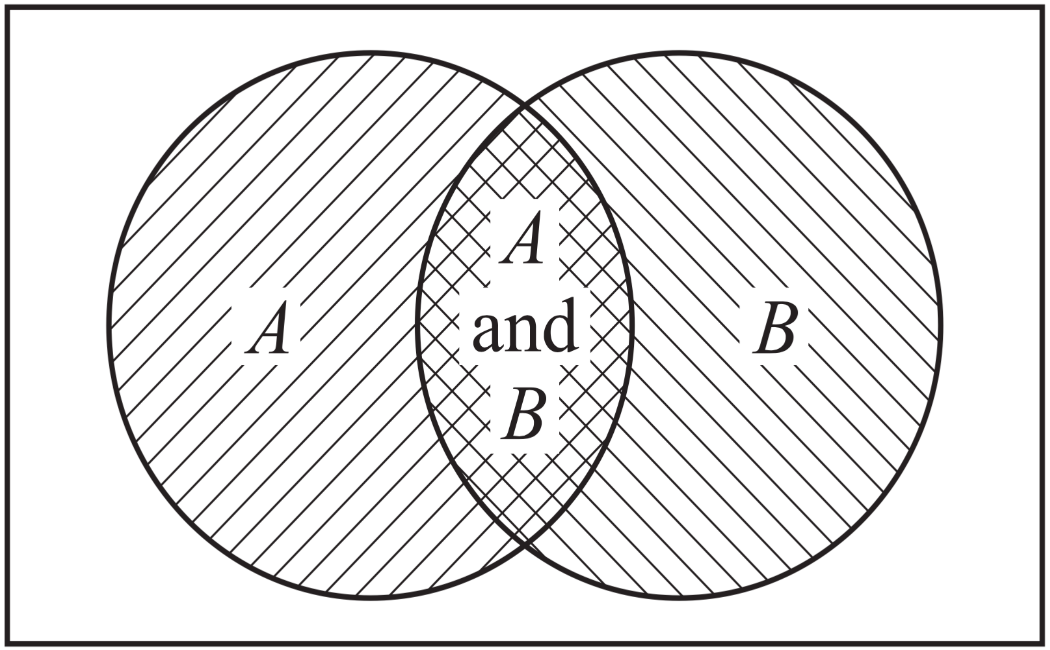
$$P(y|x) = \frac{1}{1 + e^{-w^T x}} \quad (\text{sigmoid})$$

生成式模型

通过贝叶斯定理计算每个类别的后验概率 P(y|x)，将样本分配给后验概率最大的类别。

✓ Let us look at a Venn diagram to visualize 2 events A and B. The areas of the 2 circles represent the event probabilities. The overlapping area is the event {A AND B} (i.e., $A \cap B$); the entire surface is the event {A OR B} (i.e., $A \cup B$)

1. What is the probability $P(A \cup B)$?
2. What is the probability $P(A \cap B)$?



叶斯定理韦恩图

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. We need to subtract the intersection area to avoid double-counting
- $P(A \cap B) = P(A)P(B|A)$
- Observe that we can reverse probabilities, so we also have $P(A \cap B) = P(B)P(A|B)$

► Bayes' Theorem

$$P(A|B) = P(B|A)\,{}^{P(A)}\!/_{P(B)}$$

生成式模型

$$P(y|x) = \frac{P(x|y) \cdot P(y)}{P(x)}$$

P(y)： 先验概率（每个类别出现的概率）
P(x|y)： 类条件概率密度，样本在该类下出现的概率
P(y|x)： 后验概率，已知样本后属于该类的概率

生成式模型（Generative Model） 是指通过建模数据 **x**在每一个类别 **y** 条件下的分布 **P(x|y)**，并结合先验 **P(y)**，刻画联合分布 **P(x,y)** 的模型。

模型	像谁	行为
生成式模型	记者	想知道"事情是怎么发生的"
判别式模型	法官	不管过程，只管"谁赢谁输"

生成式模型

机器学习的“生成式模型” 与大模型的“生长式AI”不同。
ChatGPT （Generative Pre-trained Transformer）
生成式预训练变换器
是能够根据输入内容生成自然语言、图像、音频等数据的 AI 模型。

ChatGPT：

$$P(x_1, x_2, \ldots, x_T) = \prod_{t=1}^{T} P(x_t | x_1, \ldots, x_{t-1})$$

是一个巨大的语言模型，逐词生成语句。

项目	机器学习的“生成式模型”	生成式 AI（ChatGPT、Stable Diffusion）
本质定义	建模 $P(x, y)$、(P(x	y))、$P(x)$
是否生成数据	✅ 是	✅ 是
是否用于分类	✅ 可以（如朴素贝叶斯）	❌ 通常不是
是否条件生成	✅ 常见	✅ 如 ChatGPT 是条件语言建模
技术复杂度	通常简单	模型极其庞大（亿级参数）

Discriminative vs. Generative (判别式vs生成式)

假设现在有一个分类问题，X是特征，Y是类标记。进一步，假设X就是两个特征（1或2），Y有两类（0或1），有如下训练样本（1，0）、（1，0）、（1，1）、（2，1）

	0	1
1	2/3	1/3
2	0	1

判别式

	0	1
1	1/2	1/4
2	0	1/4

生成式

MLE (Maximum Likelihood Estimate, 极大似然估计)

在贝叶斯分类器中，我们需要估计 P(x|y) 。但这其实就是一个概率模型的参数估计问题。那我们怎么估？极大似然估计（MLE）

The method of maximum likelihood was first introduced by Ronald Fisher, a geneticist and statistician, in the 1920s.

Most statisticians recommend this method, at least when the sample size is large, since the resulting estimators have certain desirable efficiency properties.

Ronald Fisher 1890-1962

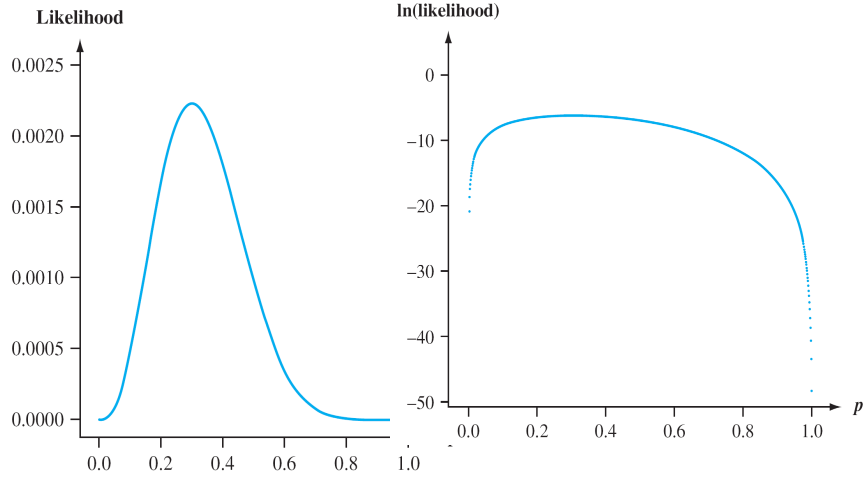
- A sample of ten new bike helmets manufactured by a certain company is obtained. Upon testing, it is found that the first, third, and tenth helmets are flawed, whereas the others are not.
- Let $p = P(\text{flawed helmet})$, i.e., p is the proportion of all such helmets that are flawed.
- Define (Bernoulli) random variables $X_1, X_2, \ldots, X_{10}$ by

$$X_1 = \begin{cases} 1 \text{ if 1st helmet is flawed} \\ 0 \text{ if 1st helmet isn't flawed} \end{cases} \quad \ldots \quad X_{10} = \begin{cases} 1 \text{ if 10th helmet is flawed} \\ 0 \text{ if 10th helmet isn't flawed} \end{cases}$$

- Then for the obtained sample, $X_1 = X_3 = X_{10} = 1$ and the other seven X_i's are all zero.
- The probability mass function of any particular X_i is $p^{x_i}(1-p)^{1-x_i}$, which becomes p if $x_i = 1$ and $1 - p$ when $x_i = 0$.
- Now suppose that the conditions of various helmets are independent of one another.
- This implies that the X_i's are independent, so their joint probability mass function is the product of the individual pmf's.

Thus the joint pmf evaluated at the observed X_i's is

$$f(x_1, \ldots, x_{10}; p) = p(1-p)p \cdots p = p^3(1-p)^7$$



Suppose that $p = .25$. Then the probability of observing the sample that we actually obtained is $(.25)^3(.75)^7 = .002086$.

If instead $p = .50$, then this probability is $(.50)^3(.50)^7 = .000977$.

For what value of p is the obtained sample most likely to have occurred? That is, for what value of p is the joint pmf as large as it can be? What value of p maximizes?

Maximum likelihood estimation （极大似然估计）

First assume a certain form of probability distribution, and then estimate parameters based on the training examples

Assuming that $P(x \mid c)$ has a certain probability distribution and is uniquely determined by the parameter θ_c ,the task is to use the training set D to estimate the parameter θ_c

The likelihood of θ_c for the set Dc composed of the cth samples in the training set D is

$$P(D_c \mid \theta_c) = \prod_{x \in D_c} P(x \mid \theta_c)$$

Multiplication is easy to cause underflow, so log-likelihood is usually used

$$LL(\theta_c) = \log\ P(D_c \mid \theta_c) = \sum_{x \in D_c} \log P(x \mid \theta_c)$$

Maximum likelihood θ_c of is estimated as $\hat{\theta}_c = \arg\max_{\theta_c} LL(\theta_c)$

The accuracy of the estimated result depends heavily on whether the assumed probability distribution form conforms to the underlying true distribution

MAP (Maximum a posteriori estimation, 极大后验估计)

In Bayesian statistics, a **maximum a posteriori probability (MAP) estimate** is an estimate of an unknown quantity, that equals the mode of the posterior distribution. The MAP can be used to obtain a point estimate of an unobserved quantity on the basis of empirical data. It is closely related to the method of maximum likehood (ML) estimation, but employs an augmented optimization objective which incorporates a prior distribution (that quantifies the additional information available through prior knowledge of a related event) over the quantity one wants to estimate. MAP estimation can therefore be seen as a regularization of maximum likelihood estimation. (from Wikipedia)

- **MLE**
  Choose θ that maximizes probability of observed data

$$\widehat{\boldsymbol{\theta}}^{\mathrm{MLE}} = \underset{\theta}{\operatorname{argmax}}\, P(Data|\theta)$$

- **Maximum a posteriori estimation (MAP)**
  Choose θ that is most probable given prior probability and data

$$\widehat{\boldsymbol{\theta}}^{\mathrm{MAP}} = \underset{\theta}{\operatorname{argmax}}\, P(\theta|D) = \underset{\theta}{\operatorname{argmax}} \frac{P(Data|\theta)P(\theta)}{P(Data)}$$

MAP估计 = 极大似然 + 先验项
loss + regularization

Bayes' theorem

一种诊断癌症的试剂，经临床试验有如下记录。得了这个癌症的人被检测出阳性的概率为 95%，未得这种癌症的人被检测出阴性的概率为 94%，而人群中得这种癌症的概率为0.5%，一个人被检测出阳性，问这个人得癌症的概率是多少？

设事件A="实验反应为阳性"，B1="被诊断者患有癌症"，B2="被诊断者不患有癌症"

$P(B1)=0.005$　　　$P(B2)=0.995$　　　$P(A|B1)=0.95$　　　$P(A|B2)=0.06$

$P(A)=P(B1)P(A|B1)+ P(B2)P(A|B2)=0.06445$

$P(B1|A)=P(B1)*P(A|B1)/P(A)=0.0737$
$P(B2|A)=P(B2)*P(A|B2)/P(A)=0.9263$

Naive Bayes Classifier （朴素贝叶斯分类器）

$$P(c \mid \boldsymbol{x}) = \frac{P(c)\boxed{P(\boldsymbol{x} \mid c)}}{P(\boldsymbol{x})}$$

Main obstacle: the joint probability of all attributes is difficult to estimate from a limited training sample

Basic idea: Assume that the attributes are independent of each other?

$$P(c \mid \boldsymbol{x}) = \frac{P(c)\,P(\boldsymbol{x} \mid c)}{P(\boldsymbol{x})} = \frac{P(c)}{P(\boldsymbol{x})}\prod_{i=1}^{d} P(x_i \mid c)$$

d is the number of attributes, xi is the value of x on the i-th attribute

$P(\boldsymbol{x})$ is the same for all categories, so

$$h_{nb}(\boldsymbol{x}) = \underset{c \in \mathcal{Y}}{\arg\max}\; P(c)\prod_{i=1}^{d} P(x_i \mid c)$$

Naive Bayes Classifier

- Estimate P(c): $P(c) = \frac{|D_c|}{|D|}$
- Estimate P(x|c):

• For discrete attributes, let D_{c,x_i} denote the set of samples in Dc whose value is xi on the i-th attribute, then

$$P(x_i \mid c) = \frac{|D_{c,x_i}|}{|D_c|}$$

• For continuous attributes, consider the probability density function, assuming $p(x_i \mid c) \sim \mathcal{N}(\mu_{c,i}, \sigma_{c,i}^2)$

$$p(x_i \mid c) = \frac{1}{\sqrt{2\pi}\sigma_{c,i}} \exp\left(-\frac{(x_i - \mu_{c,i})^2}{2\sigma_{c,i}^2}\right)$$

下面我们用西瓜数据集 3.0 训练一个朴素贝叶斯分类器，对测试例“测 1”进行分类：

编号	色泽	根蒂	敲声	纹理	脐部	触感	密度	含糖率	好瓜
测 1	青绿	蜷缩	浊响	清晰	凹陷	硬滑	0.697	0.460	?

首先估计类先验概率 $P(c)$，显然有

$$P(\text{好瓜}=\text{是}) = \frac{8}{17} \approx 0.471\ ,$$
$$P(\text{好瓜}=\text{否}) = \frac{9}{17} \approx 0.529\ .$$

$$P(\text{好瓜}=\text{是}) \times P_{\text{青绿}|\text{是}} \times P_{\text{蜷缩}|\text{是}} \times P_{\text{浊响}|\text{是}} \times P_{\text{清晰}|\text{是}} \times P_{\text{凹陷}|\text{是}}$$
$$\times P_{\text{硬滑}|\text{是}} \times p_{\text{密度: 0.697}|\text{是}} \times p_{\text{含糖: 0.460}|\text{是}} \approx 0.038\ ,$$
$$P(\text{好瓜}=\text{否}) \times P_{\text{青绿}|\text{否}} \times P_{\text{蜷缩}|\text{否}} \times P_{\text{浊响}|\text{否}} \times P_{\text{清晰}|\text{否}} \times P_{\text{凹陷}|\text{否}}$$
$$\times P_{\text{硬滑}|\text{否}} \times p_{\text{密度: 0.697}|\text{否}} \times p_{\text{含糖: 0.460}|\text{否}} \approx 6.80 \times 10^{-5}\ .$$

由于 $0.038 > 6.80 \times 10^{-5}$，因此，朴素贝叶斯分类器将测试样本“测 1”判别为“好瓜”.

需注意，若某个属性值在训练集中没有与某个类同时出现过，则直接基于式(7.17)进行概率估计，再根据式(7.15) 进行判别将出现问题. 例如，在使用西瓜数据集 3.0 训练朴素贝叶斯分类器时，对一个“敲声=清脆”的测试例，有

$$P_{\text{清脆}|\text{是}} = P(\text{敲声}=\text{清脆} \mid \text{好瓜}=\text{是}) = \frac{0}{8} = 0\ ,$$

然后，为每个属性估计条件概率 $P(x_i \mid c)$：

$$P_{\text{青绿}|\text{是}} = P(\text{色泽}=\text{青绿} \mid \text{好瓜}=\text{是}) = \frac{3}{8} = 0.375\ ,$$
$$P_{\text{青绿}|\text{否}} = P(\text{色泽}=\text{青绿} \mid \text{好瓜}=\text{否}) = \frac{3}{9} \approx 0.333\ ,$$
$$P_{\text{蜷缩}|\text{是}} = P(\text{根蒂}=\text{蜷缩} \mid \text{好瓜}=\text{是}) = \frac{5}{8} = 0.375\ ,$$
$$P_{\text{蜷缩}|\text{否}} = P(\text{根蒂}=\text{蜷缩} \mid \text{好瓜}=\text{否}) = \frac{3}{9} \approx 0.333\ ,$$
$$P_{\text{浊响}|\text{是}} = P(\text{敲声}=\text{浊响} \mid \text{好瓜}=\text{是}) = \frac{6}{8} = 0.750\ ,$$
$$P_{\text{浊响}|\text{否}} = P(\text{敲声}=\text{浊响} \mid \text{好瓜}=\text{否}) = \frac{4}{9} \approx 0.444\ ,$$
$$P_{\text{清晰}|\text{是}} = P(\text{纹理}=\text{清晰} \mid \text{好瓜}=\text{是}) = \frac{7}{8} = 0.875\ ,$$
$$P_{\text{清晰}|\text{否}} = P(\text{纹理}=\text{清晰} \mid \text{好瓜}=\text{否}) = \frac{2}{9} \approx 0.222\ ,$$
$$P_{\text{凹陷}|\text{是}} = P(\text{脐部}=\text{凹陷} \mid \text{好瓜}=\text{是}) = \frac{6}{8} = 0.750\ ,$$
$$P_{\text{凹陷}|\text{否}} = P(\text{脐部}=\text{凹陷} \mid \text{好瓜}=\text{否}) = \frac{2}{9} \approx 0.222\ ,$$
$$P_{\text{硬滑}|\text{是}} = P(\text{触感}=\text{硬滑} \mid \text{好瓜}=\text{是}) = \frac{6}{8} = 0.750\ ,$$
$$P_{\text{硬滑}|\text{否}} = P(\text{触感}=\text{硬滑} \mid \text{好瓜}=\text{否}) = \frac{6}{9} \approx 0.667\ ,$$
$$p_{\text{密度: 0.697}|\text{是}} = p(\text{密度}=0.697 \mid \text{好瓜}=\text{是}) = \frac{1}{\sqrt{2\pi}\cdot 0.129}\exp\left(-\frac{(0.697-0.574)^2}{2\cdot 0.129^2}\right) \approx 1.959\ ,$$
$$p_{\text{密度: 0.697}|\text{否}} = p(\text{密度}=0.697 \mid \text{好瓜}=\text{否}) = \frac{1}{\sqrt{2\pi}\cdot 0.195}\exp\left(-\frac{(0.697-0.496)^2}{2\cdot 0.195^2}\right) \approx 1.203\ ,$$
$$p_{\text{含糖: 0.460}|\text{是}} = p(\text{含糖率}=0.460 \mid \text{好瓜}=\text{是}) = \frac{1}{\sqrt{2\pi}\cdot 0.101}\exp\left(-\frac{(0.460-0.279)^2}{2\cdot 0.101^2}\right) \approx 0.788\ ,$$
$$p_{\text{含糖: 0.460}|\text{否}} = p(\text{含糖率}=0.460 \mid \text{好瓜}=\text{否}) = \frac{1}{\sqrt{2\pi}\cdot 0.108}\exp\left(-\frac{(0.460-0.154)^2}{2\cdot 0.108^2}\right) \approx 0.066\ .$$

Laplacian correction

- If a certain attribute value does not appear in the training set at the same time as a certain class, there will be problems in the direct calculation, because the probability multiplication will "erase" the information provided by other attributes

For example, if there is no good melon with "knock = crisp" in the training set, the model will encounter the test sample of "knock = crisp"...

- Let N denote the number of possible categories in the training set D, and Ni denote the number of possible values of the i-th attribute

Assuming a uniform distribution of attribute values and categories, this is an additional bias introduced

类别先验概率

$$\hat{P}(c) = \frac{|D_c| + 1}{|D| + N}$$

N是类别个数

条件概率

$$\hat{P}(x_i|c) = \frac{|D_{c,x_i}| + 1}{|D_c| + N_i}$$

Ni是第i个属性的可能取值个数

7.3 试编程实现拉普拉斯修正的朴素贝叶斯分类器，并以西瓜数据集3.0为训练集，对p151“测1”样本进行判别